

2026-05-01 16:03:31

Tickers: AAL, AAPL, ADBE, AMAT, AMC, AMD, AMZN, AVGO, BA, BABA, BAC, C, COIN, COST, CRWD, CVNA, CVS, F, GLD, GOOGL, HOOD, IBIT, INTC, IWM, JPM, KO, KRE, LLY, MARA, META, MRVL, MSFT, MSTR, MU, NFLX, NIO, NVDA, ORCL, PFE, PLTR, QQQ, RIOT, SBUX, SLV, SMC, SOFI, SPOT, SPY, TSLA, TSM, UBER, UNH, UPS, USO, V, VXX, WMT, XBI, XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XOM, XOP

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[{"type": "markdown", "content": "# OI Concentration Signals: Per-Ticker Predictive Scan## `pct_oi_within_5pct_co` & `pct_oi_within_10pct_co` \u2192 Forward Returns (1d & 5d)\n\n### Executive Summary\n\nThis scan evaluates whether the percentage of open interest concentrated near the current spot price (within 15% and 110% of close) carries predictive information for 1-day and 5-day forward returns across a broad ticker universe. The short answer: yes, but selectively. Signal quality is highly ticker-specific, the dominant horizon is 5-day (ret_5d_fwd_oc), and the most reliable signals manifest via spot_co (a composite of OI concentration at spot) and its directional decompositions (pct_oi_above_spot_co, pct_oi_below_spot_co) rather than the raw concentration variants directly \u2014 though pct_oi_within_5pct_co does surface independently for TSLA (score 74).\n\nImportantly, OI-based metrics here are treated as OI-implied gamma proxies \u2014 OI does not reveal dealer inventory, so these signals reflect structural positioning concentration, not confirmed dealer hedging flows."}, {"type": "chart", "chart_id": "ticker_scoreboard", "title": "Best Composite Score by Ticker", "config": {"type": "bar", "data": {"labels": ["AAL", "USO", "V", "TSLA", "XLB", "XLP", "COIN", "AAPL", "C", "XOP", "VXX", "XLV", "BA", "SPY", "RIOT", "WMT", "CRWD", "SOFI", "KO", "COST", "META", "LLY", "MSTR", "SBUX", "XLE", "AMD", "AMZN", "KRE", "BABA", "CVS", "UNH", "F", "QQQ", 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The non-monotonic mid-range weakens the IC but the tail behavior is real and consistent.\n\n**TSLA (score 78)** mirrors AAL structurally \u2192 Buckets 1\u2192132 generate +5.0% and +3.0% respectively \u2192 but the middle buckets are noisier, which is characteristic of a high-vol name. `pct\_oi\_within\_5pct\_co` specifically scores 74 for TSLA, confirming that tightly-concentrated OI near spot is a relevant regime indicator for this ticker."}, {"type": "chart", "chart_id": "bucket_AAL_spot_co_ret_5d_fwd_oc", "title": "Decile Profile: AAL | spot_co \u2192 ret_5d_fwd_oc (score=80)", "config": {"type": "bar", "data": {"labels": ["D1", "D2", "D3", "D4", "D5", "D6", "D7", "D8", "D9", "D10"], "datasets": [{"label": "Avg Return %", "data": [3.9903, 1.7314, 0.291, -0.202, 0.1596, -0.8107, -1.2884, -0.4347, -2.0898, -2.1949], "backgroundColor": ["#3498db", "#3498db", "#3498db", "#e84393", "#3498db", "#e84393", "#e84393", "#e84393", "#e84393", "#e84393"], "borderWidth": 0}}, {"responsive": 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This is a *positive* relationship \u2192 2014 more OI sitting above spot predicts upward price drift, consistent with a call-wall pinning or dealer-long-delta interpretation. The equity curve for the top decile compounds to +58.2% with a modest -14.4% max drawdown. Note: `pct\_o\_i\_below\_spot\_co` also scores 75 for V, with the inverse logic (linear-weak-negative), confirming the directional coherence.\n\n**XLB (score 76)** and **XLP (score 76)** are sector ETFs with structurally cleaner signals than single names. XLB's `spot\_co` profile is monotonic through Buckets 1 \u2192 20139 with low noise. XLP's `pct\_o\_i\_below\_spot\_co` signal (score 76) and `pct\_o\_i\_above\_spot\_co` (score 74) are complementary \u2192 2014 together they suggest XLP OI structure is informationally rich. **XLV** (score 74) adds a third sector ETF with threshold behavior on both above/below decompositions."}, {"type": "chart", "chart_id": "bucket_V_pct_o_i_above_spot_co_ret_5d_fwd_oc", "title": "Decile Profile: V | pct_o_i_above_spot_co \u2192 ret_5d_fwd_oc (score=78)", "config": {"type": "bar", "data": {"labels": ["D1", "D2", "D3", "D4", "D5", "D6", "D7", "D8", "D9", "D10"], "datasets": [{"label": "Avg Return %", "data": [-0.4214, -0.0853, 0.2768, -0.316, -0.2969, -0.0117, 0.3668, 0.5008, 0.6447, 1.4372], "backgroundColor": ["#e84393", "#e84393", "#3498db", "#e84393", "#e84393", "#e84393", "#e84393", "#3498db", "#3498db", "#3498db"], "borderWidth": 0}], "options": {"responsive": true, "maintainAspectRatio":

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AAPL's dual signal (both `pct\_oi\_above` and `pct\_oi\_below` score 75) with concentration of 0.25 suggests the signal is real but regime-dependent. **BA (score 73)** echoes the AAL linear-negative pattern. **C (score 75)** via `put\_call\_oi\_ratio` and **VXX (score 74)** via the same signal are adjacent signals that suggest OI ratio structure matters for financials and vol instruments.\n\nThe **multi-factor interaction** testing reveals marginal lift: AAPL's above+below combo produces a lift of -0.33 (i.e., the interaction reduces the signal) and XLB's spot_co+rv_20d combo shows that the OI signal is *amplified in

[illegible]

29.8, 43.0, 48.1, 35.4, 36.0, 42.4, 42.4, 55.7, 44.9, 48.1, 57.0, 55.7, 40.7, 42.9, 48.1, 48.1, 52.5, 61.4, 37.3, 41.3, 43.1, 46.0, 43.0, 53.8, 50.0, 57.0, 48.7, 42.4, 46.8, 39.9, 37.3, 47.5, 52.5, 38.0, 38.0, 63.3, 56.4, 44.3, 44.9, 72.2, 43.7, 51.9, 46.2, 40.5, 34.2, 38.0, 60.1, 31.0, 62.5, 45.6, 47.5, 56.3, 61.4, 55.7, 54.4, 44.3, 43.7, 43.0, 44.3, 47.5], "backgroundColor": "#3498db", "borderWidth": 0}, {"label": "D1 Win Rate %", "data": [69.8, 39.0, 62.3, 59.1, 37.1, 49.1, 55.4, 48.4, 66.0, 55.4, 59.1, 58.5, 65.1, 61.6, 73.0, 35.9, 66.7, 53.5, 59.1, 65.4, 73.1, 61.1, 43.4, 60.4, 54.1, 39.6, 62.9, 53.3, 53.2, 61.6, 61.6, 70.4, 50.9, 61.6, 62.3, 61.0, 70.4, 58.5, 62.9, 57.1, 71.7, 56.0, 69.2, 48.4, 43.3, 66.7, 57.2, 55.4, 65.4, 48.4, 60.4, 59.1, 64.8, 61.0, 50.3, 40.2, 57.1, 61.6, 57.9, 56.6, 46.5, 60.4, 64.1, 59.8, 63.5, 68.5, 50.3, 55.4], "backgroundColor": "#e84393", "borderWidth": 0}}, {"options": {"responsive": true, "maintainAspectRatio": false, "animation": false, "plugins": {"legend": {"labels": {"color": "#aaa", "font": {"size": 10}}}, "tooltip": {"backgroundColor": "rgba(20,20,20,0.95)", "borderColor": "#444", "borderWidth": 1}, "scales": {"x": {"ticks": {"color": "#888", "font": {"size": 9}, "maxRotation": 45}, "grid": {"color": "rgba(255,255,255,0.05)", "border": {"color": "transparent"}}, "y": {"ticks": {"color": "#888", "font": {"size": 9}, "grid": {"color": "rgba(255,255,255,0.05)", "border": {"color": "transparent"}}}}}, {"type": "markdown", "content": "### What Did NOT Work\n\n- **1-day horizon**: No ticker in this scan produces a material IC at the 1-day horizon for OI concentration signals. The signal is structurally a 5-day phenomenon, likely because OI concentration influences pin/magnet dynamics over a multi-day window, not intraday.\n\n- **Raw `pct\_oi\_within\_5pct\_co` and `pct\_oi\_within\_10pct\_co` as standalone signals**: These score well only for TSLA (74) among the direct variants. Most tickers require the directional decomposition (`above` vs. `below` spot) or the composite `spot\_co` to achieve t-stat significance. The raw symmetric concentration metric is largely uninformative in isolation.\n\n- **SPY**: The top signal for SPY (score 72, `weighted\_avg\_dte`) is a DTE signal, not an OI concentration signal \u2014 confirming that the OI structure metrics have limited edge on the most liquid, most efficiently-priced underlying.\n\n### Actionable Conclusions\n\n1. **Best single implementation**: Long bottom-decile AAL when `spot\_co` is lowest (high OI concentration below spot), targeting 5-day exit. 100% yearly consistency, +239.5% cumulative, -26% max DD.\n\n2. **Most structurally clean long signal**: V top-decile `pct\_oi\_above\_spot\_co` \u2014 2014 monotonic, low drawdown (+58.2% cum, -14.4% max DD), easily implementable as a weekly rebalance.\n\n3. **Sector ETF overlay**: XLB and XLP OI concentration signals provide sector-level conditioners \u2014 use as regime filters when positioning in Materials and Staples.\n\n4. **Volatility conditioning**: For XLB specifically, restrict OI concentration signals to elevated RV regimes (rv_20d high) for incremental win rate lift.\n\n5. **Do not extend to 1-day**: The 1-day horizon shows no reliable signal. Minimum holding period for any OI concentration-derived trade is 5 trading days."}, {"type": "markdown", "content": "---\n\n## Caveats & Challenges\n\n## Skeptic Review\n\n### 1. Sample Size and Precision Claims\n\nMinimum bucket size of 158 observations across 10 deciles from ~1,585 total observations is borderline acceptable but not comfortable. The equity curve statistics are the more concerning figures: AAL's top-decile short has only **41 trades** over 7 years; USO's bottom-decile long has **34 trades**. Claiming a '+239.5% cumulative return' and '100% consistency' from 34\u20141350 trades is statistically thin. Standard errors on win rates of 70% with n=50 are 0.00b113% at 95% confidence \u2014 meaning the 'true' win rate plausibly extends down to 57%, which would materially compress the Sharpe claims.\n\n### 2. Regime Concentration Risk\n\n'100% yearly consistency over 7 years' sounds impressive but deserves scrutiny. The concentration metric for AAL is 0.37 \u2014 the report defines this but doesn't explain what fraction of the total P&L is attributable to 2020\u20142021, when AAL underwent bankruptcy-adjacent volatility, USO underwent a structural reconstitution (negative oil prices), and TSLA had unprecedented gamma events. Signals with concentration scores above 0.30 in volatile single-names are regime-fragile by construction. The consistency claim needs a breakdown of *magnitude* of returns per year, not just directional accuracy.\n\n### 3. Monotonicity vs. Tail Concentration\n\nThe report acknowledges USO's mid-range is 'non-monotonic' and TSLA's middle buckets are 'noisier,' yet both receive Tier 1 or Tier 2 designations. The mid-decile range for USO (0.0156) and AAL (0.0145) are non-trivial, but if the bulk of the bucket spread is driven by Buckets 1 and 10 while Buckets 3\u2014138 are flat or noisy, the IC is being flattered by tail events rather than structural linearity. A score of 80 built on two extreme deciles is not the same as a score of 80 built on a smooth 10-bucket ramp.\n\n### 4. Drawdown and Tradeability\n\nAAL's top-decile strategy has a **max drawdown of -129.7%** \u2014 2014 which exceeds 100% and implies leveraged or compounded losses that would have wiped out capital before recovering. USO's top-decile max drawdown is **-149.5%**. These are not tradeable strategies without position sizing so conservative that the reported cumulative returns become largely irrelevant. The -126.9% cumulative return on AAL top-decile is further complicated by USO's structural 2020 roll crisis. No transaction cost or bid-ask adjustment is disclosed; for AAL and USO

\u2014 both liquidity-challenged at times \u2014 slippage on a 5-day hold in options-adjacent names can be material.\n\n### 5. OI Proxy Limitations\n\nThe report correctly flags that OI does not reveal dealer inventory and uses \"OI-implied gamma proxy\" language. However, the narrative then interprets V's result as \"call-wall pinning or dealer-long-delta\" behavior \u2014 which reintroduces the causal dealer framing that the disclaimer was meant to prevent. The signal may be real; the mechanism is speculative and untestable from OI data alone.\n\n### 6. No Multiple Testing Correction\n\nThe scan covers a \"broad\"]